

Question 1: Simple Loop

What is the asymptotic running time of the following pseudocode?

```
1 for i = 1 to n
2     print("Hello")
```

Question 2: Nested Loops

Analyze the time complexity of the following code:

```
1 for i = 1 to n
2     for j = 1 to i
3         print(i * j)
```

Question 3: Logarithmic Loop

Find the running time of:

```
1 i = 1
2 while i ≤ n
3     print(i)
4     i = i * 2
```

Question 4: Prove Big-O

Prove that:

$$3n^2 + 7n + 5 = O(n^2)$$

Question 5: Prove Big-Ω

Prove that:

$$5n \log n + 20n = \Omega(n \log n)$$

Question 6: Prove Θ Bound

Show that:

$$7n^3 + 3n^2 + 10 = \Theta(n^3)$$